

FAKE NEWS DETECTION USING MACHINE LEARNING

MINOR PROJECT REPORT

Submitted by

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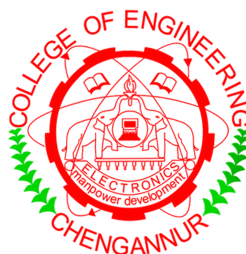
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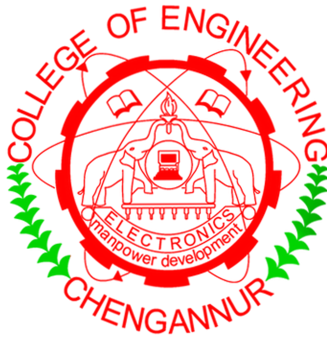
*in partial fulfillment of the requirements for the award of B.Tech in
Electronics and Communication Engineering with minor in **B.Tech**
Computer Science and Engineering*



**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
COLLEGE OF ENGINEERING CHENGANNUR, ALAPPUZHA**

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**DEPARTMENT OF COMPUTER ENGINEERING
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CERTIFICATE

*This is to certify that the project report titled **Fake News Detection Using Machine Learning** is a bonafide record of the **CSD481 MINI PROJECT** presented by **Nandana V D (CHN22EE032)**, **Aswathy Krishnan(CHN22EC032)**, **Jobin Abraham Chacko (CHN22EC053)**, **K M Karthika Krishna (CHN22EC062)**, Seventh Semester B.Tech Degree student, under my guidance and supervision. This project is submitted in partial fulfillment of the requirements for the award of **B.Tech Degree in Electronics and Communication with minor in B.Tech Computer Science and Engineering** of **APJ Abdul Kalam Technological University**.*

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DECLARATION

We undersigned hereby declare that the project report “**Fake News Detection Using Machine Learning**” , submitted for partial fulfillment of the requirements for the award of B.Tech Degree of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under the supervision of **Dr. Geetha S**, Associate Professor, Department of Computer Engineering. This submission represents our ideas in our own words, and where ideas or words of others have been included, We have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma, or similar title of any other University.

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ABSTRACT

In today's digital era, the exponential growth of online news and social media has made information sharing faster and more accessible than ever. However, this has also accelerated the spread of false or misleading information, commonly known as fake news, which can manipulate public perception, influence political and economic decisions, and harm social harmony. This project presents a machine learning-based framework for detecting fake news using the ISOT Fake and Real News Dataset, which contains over 44,000 labeled news articles. The system involves text preprocessing steps such as cleaning, stopword removal, and lemmatization, followed by feature extraction through the Term Frequency–Inverse Document Frequency (TF-IDF) technique. Logistic Regression is employed as the core classifier due to its efficiency and proven performance in text classification tasks. The model is evaluated using standard metrics, including accuracy, precision, recall, and F1-score, to ensure balanced performance.

The experimental results demonstrate that the Logistic Regression model, combined with TF-IDF features, achieves an accuracy between 92% and 94%, indicating its strong ability to distinguish between real and fake news articles. The study highlights that even a simple and computationally efficient machine learning pipeline can yield competitive performance. In conclusion, this work emphasizes the effectiveness of machine learning in combating digital misinformation and restoring credibility in online information systems. The developed framework provides a scalable and reliable approach that can be integrated into real-world fact-checking and media verification platforms.

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Chapter 1

INTRODUCTION

1.1 OVERVIEW

In the digital age, online news and social media platforms have revolutionized the way people consume information. Millions of articles, posts, and updates are published daily, allowing users to access news instantly from anywhere in the world. However, this convenience has also introduced a major challenge—the rapid spread of fake news. Fake news refers to deliberately fabricated or misleading information presented as legitimate news, often created to influence opinions, manipulate political or social narratives, or generate financial gain through sensationalism. The prevalence of such misinformation can lead to serious consequences, including public confusion, social unrest, and erosion of trust in credible media sources. Manual methods of verifying the authenticity of news are neither efficient nor scalable, given the overwhelming volume of digital content generated every second. Traditional fact-checking organizations depend on human experts, who take time to validate sources, cross-reference facts, and assess linguistic cues. Although accurate, this process is slow and subjective. To overcome these limitations, researchers and technologists are increasingly turning to Machine Learning (ML) and Natural Language Processing (NLP) techniques to automate the detection of fake news based on data-driven insights extracted from textual content.]

1.2 Project Area

The project “Fake News Detection Using Machine Learning” belongs to the interdisciplinary domain of Artificial Intelligence (AI), primarily focusing on Machine Learning (ML) and Natural Language Processing (NLP). It aims to develop an intelligent system capable of automatically identifying and classifying news articles as real or fake based on their textual content. The project integrates multiple areas of computing such as data science, text analytics, and information security. Machine Learning provides the foundation for building predictive models that learn from historical data and detect patterns that distinguish truthful news from fabricated stories. NLP techniques are applied to process and clean textual data through tokenization, stop-word removal, lemmatization, and vectorization using methods like Term Frequency–Inverse Document Frequency (TF-IDF). These steps transform unstructured text into a numerical format suitable for model training. The

project also falls within the broader scope of data science, involving data preprocessing, exploration, visualization, and evaluation using performance metrics such as accuracy, precision, recall, and F1-score. Moreover, it contributes to the field of cybersecurity and information integrity by addressing the growing concern of misinformation and its impact on public trust. Overall, this project area combines the principles of artificial intelligence, language processing, and data analytics to build a reliable system that supports truth verification and promotes the responsible use of digital information.

1.3 Objectives

The main objective of the project “Fake News Detection Using Machine Learning” is to develop an intelligent and automated system capable of distinguishing between real and fake news articles using machine learning techniques. The project aims to design an efficient framework that processes textual data, extracts meaningful linguistic features, and accurately classifies news content based on authenticity. One of the key goals is to apply Natural Language Processing (NLP) techniques such as tokenization, stop-word removal, lemmatization, and TF-IDF vectorization to convert unstructured text into machine-readable form. The project also seeks to implement and evaluate machine learning algorithms, with a focus on Logistic Regression, to achieve high accuracy, precision, and recall in detecting misinformation. In addition, the system is intended to serve as a baseline model that can be extended to more advanced deep learning architectures in future research. Through this approach, the project contributes to building reliable digital ecosystems, promoting trust in online information, and providing a scalable solution for mitigating the harmful effects of fake news dissemination across social and digital media platforms.

1. To develop an automated machine learning model that can accurately classify news articles as real or fake using the ISOT Fake and Real News dataset.
2. To apply Natural Language Processing (NLP) techniques such as text cleaning, tokenization, stop-word removal, lemmatization, and TF-IDF vectorization for efficient feature extraction from raw text data
3. To evaluate and validate the performance of the proposed model using key metrics like accuracy, precision, recall, and F1-score, and establish a reliable baseline for future enhancement using advanced deep learning methods.

Chapter 2

PROBLEM DEFINITION AND MOTIVATIONS

The rapid growth of online news and social media has made it increasingly difficult to distinguish between authentic and fabricated information. Fake news, which refers to intentionally false or misleading content presented as factual news, spreads rapidly across digital platforms and influences public opinion, political outcomes, and social harmony. Manual verification of such content is slow, subjective, and ineffective when dealing with the vast volume of data generated every second. Therefore, there is a critical need for an automated and reliable system that can accurately detect fake news using data-driven methods. The problem addressed in this project is to design and implement a machine learning-based framework that can process textual data, extract meaningful linguistic and statistical features, and classify news articles as real or fake. The proposed system aims to enhance information credibility by providing a scalable and efficient solution capable of real-time detection, thereby reducing the impact of misinformation in the digital ecosystem.

2.1 Existing System

In the existing systems, fake news detection primarily relies on manual verification and rule-based methods, which involve human intervention to evaluate the authenticity of news content. Traditional approaches depend on fact-checking organizations and media agencies that manually review articles, verify sources, and cross-check claims against credible databases. Although these methods can be accurate, they are time-consuming, labor-intensive, and not scalable for the massive amount of information shared online every day. Some earlier automated systems used keyword-based filtering or simple statistical models to identify suspicious content. These systems focused on specific words or phrases frequently associated with fake news, but such methods often fail when fake news uses neutral or factual language. Additionally, traditional algorithms lacked the ability to understand linguistic context, sarcasm, or the subtle manipulation of facts, which are common characteristics of modern misinformation. Another limitation of older systems is their dependency on limited datasets and manual feature engineering, which restrict their generalization ability to unseen or diverse news domains. As a result, the accuracy and reliability of these systems remain low, especially when applied to real-world data with varied writing styles and topics. Overall, existing systems struggle to handle the complexity, diversity, and high volume of online news data. This creates a strong mo-

tivation for developing more advanced, machine learning–based approaches that can automatically learn patterns, adapt to changing trends, and provide faster and more accurate fake news detection.

2.2 Limitations

- **Limited Generalization to New Domains** The model is trained on the ISOT dataset, which contains news articles from specific sources. When exposed to news from different platforms or writing styles (e.g., social media posts, blogs), its accuracy may decrease due to domain shift.
- **Dependence on Textual Features Only** The system relies solely on the text content of news articles. It does not consider other useful indicators such as author credibility, source reputation, publication time, or metadata — all of which could improve prediction reliability.
- **Lack of Contextual Understanding** Traditional ML models like Logistic Regression with TF-IDF only analyze word frequencies and ignore semantic relationships or contextual meanings. As a result, they may misclassify sarcastic or nuanced statements.

2.3 Problem Statement

The increasing spread of fake news across digital media platforms has become a serious challenge, influencing public opinion, politics, and social stability. Manual verification of news articles is time-consuming, subjective, and inefficient for handling the vast amount of online information generated every day. Therefore, there is a growing need for an automated and reliable system that can effectively distinguish between fake and real news. This project aims to develop a machine learning-based model that analyzes the textual content of news articles to identify deceptive information. By applying natural language processing (NLP) techniques for data cleaning and feature extraction, and training classification algorithms such as Logistic Regression, the system seeks to provide a scalable and accurate solution for fake news detection. This project focuses on building a machine learning model that can analyze news articles, extract meaningful textual features, and classify them as real or fake. The aim is to develop an effective, scalable, and data-driven approach to combat misinformation and support digital media integrity.

2.4 Proposed System

The proposed system aims to automate the process of detecting fake news by using machine learning techniques applied to textual data. The system follows a structured pipeline consisting of data collection, preprocessing, feature extraction, model training, and evaluation. News articles are first gathered from a reliable dataset containing both real and fake samples. The text data is then preprocessed by converting all words to lowercase, removing punctuation, stopwords, and irrelevant characters, and performing tokenization and lemmatization to normalize the text. After preprocessing, the textual information is converted into numerical form using the TF-IDF (Term Frequency–Inverse Document Frequency) vectorization technique, which effectively captures the importance of words within each document. These extracted features are then fed into a supervised machine learning model, specifically Logistic Regression, which learns to distinguish between fake and real news based on the patterns identified in the training data. The trained model is evaluated using performance metrics such as Accuracy, Precision, Recall, and F1-score to measure its effectiveness. This system provides a simple yet powerful baseline approach for automated fake news detection, ensuring efficient and scalable classification of online information. It can later be extended with deep learning models or real-time deployment for enhanced accuracy and broader applicability.

2.4.1 System Architecture

Data Collection: The system uses the ISOT Fake News Dataset, which contains labeled fake and real news articles.

Data Preprocessing: Raw text data often includes noise such as punctuation, numbers, and irrelevant symbols. These are removed to ensure consistency. Stopwords are eliminated, and words are converted to lowercase to reduce redundancy. Lemmatization is then applied to transform words into their base forms.

Feature Extraction: Once cleaned, the text is converted into numerical vectors using the TF-IDF (Term Frequency–Inverse Document Frequency) technique. This step captures the importance of words relative to the dataset, enabling the model to differentiate between fake and real news based on linguistic patterns.

Model Training and Testing: The processed data is split into training and testing subsets. A Logistic Regression model is trained on the TF-IDF vectors, learning to identify patterns that signify fake or real news. The trained model is then tested to evaluate its predictive performance.

2.4.2 System Workflow

The workflow of the proposed system follows a step-by-step process for efficient fake news classification: Input Stage: News articles are provided as input from the dataset.

Preprocessing Stage: The text undergoes cleaning, tokenization, and lemmatization to prepare it for vectorization.

Vectorization Stage: The TF-IDF vectorizer converts the text into a numerical representation that the model can interpret.

Training Stage: The Logistic Regression classifier is trained on the feature vectors to learn the distinction between fake and real news.

Evaluation Stage: The model's performance is assessed using metrics such as Accuracy, Precision, Recall, and F1-score.

Output Stage: The final prediction label—either “Fake” or “Real”—is generated for each news article.

The proposed system is designed to automatically detect fake news using a machine learning-based approach that combines natural language processing and text classification techniques. The system begins with data collection, where real and fake news articles are gathered from the ISOT dataset. The collected text data then undergoes preprocessing to remove noise such as punctuation, stopwords, and special characters, followed by converting all words to lowercase and applying lemmatization to normalize them. After preprocessing, the clean text is transformed into numerical form using the TF-IDF (Term Frequency–Inverse Document Frequency) method, which represents the importance of each word relative to the dataset. These numerical features are then used to train a Logistic Regression model that learns to differentiate between fake and real news articles. The trained model is tested using a separate dataset and evaluated with metrics such as Accuracy, Precision, Recall, and F1-score to ensure reliability. This systematic workflow allows the model to effectively classify news articles and provides a scalable baseline that can be further enhanced using advanced deep learning models or real-time implementation in future work.

Chapter 3

LITERATURE REVIEW

Fake news detection has become a significant research area due to the rapid spread of misinformation through digital platforms and social media. Researchers have explored various machine learning and deep learning techniques to improve the accuracy and reliability of automated fake news classification systems. Traditional machine learning algorithms such as Logistic Regression, Naive Bayes, Decision Trees, Random Forest, and XGBoost have been widely applied using textual feature extraction methods like TF-IDF. More recently, transformer-based and hybrid models have also gained attention for their contextual understanding capabilities.

Uma Sharma, Sidarth Saran, and Shankar M. Patil (2021) conducted **a comparative analysis of machine learning algorithms** for fake news detection using the LIAR dataset consisting of 12.8k labeled statements [1]. The study evaluated Logistic Regression, Naive Bayes, Decision Tree, and Random Forest models. Their findings indicated that Logistic Regression combined with TF-IDF features achieved the highest accuracy, highlighting the effectiveness of simple and interpretable models in text classification tasks.

Biplob Kumar Sutradhar et al. (2023) developed **a fake news detection system using a custom dataset of approximately 1,800 news articles** [2]. The authors compared Naive Bayes and Logistic Regression classifiers. The results showed that Naive Bayes achieved the best accuracy of 56 percent, emphasizing that model performance is strongly influenced by dataset size and diversity.

Naveed Sheikh et al. (2024) performed **an empirical study comparing Logistic Regression and Naive Bayes on a dataset of 8,900 news items** [3]. The research demonstrated that Logistic Regression significantly outperformed Naive Bayes, achieving an accuracy of 98 percent, thereby confirming its robustness for text-based fake news classification.

Vyankatesh Rampurkar and Thirupurasundari D.R. (2024) applied **classical machine learning models on the ISOT dataset using TF-IDF features** [4]. Their experimental results identified Logistic Regression as an effective and computationally efficient approach for binary fake news detection tasks.

Omar Bashaddadh et al. (2025) presented **a comprehensive review of machine learning and deep learning techniques for fake news detection** [5]. The study analyzed 90 peer-reviewed arti-

cles published between 2020 and 2024. The review highlighted the dominance of transformer-based models while also emphasizing challenges related to dataset quality, generalization, and deployment strategies.

Alaa Altheneyan and Aseela Alhadlaq (2023) proposed a **distributed fake news detection system using Apache Spark and a stacked ensemble learning approach** [6]. Evaluated on the FNC-1 dataset, the model achieved an F1-score of 92.45, demonstrating scalability and efficiency for large-scale fake news classification.

Akanbi Caleb et al. (2025) investigated **hybrid models combining BERT and XGBoost for fake news detection using the LIAR dataset** [7]. Their findings revealed that the hybrid model achieved an accuracy of 73, suggesting that integrating contextual embeddings with traditional classifiers can enhance performance.

Anwar V. Mbaziira (2024) introduced an **explainable fake news detection model using XGBoost along with linguistic and psycholinguistic features** [8]. The approach improved both classification accuracy and interpretability, addressing one of the major challenges in machine learning-based misinformation detection systems.

S.A. Al-Obaidi (2024) focused on **handling dataset imbalance in fake news detection using the XGBoost algorithm on the FakeNewsNet dataset** [9]. The study proposed techniques to mitigate class imbalance, resulting in improved model stability and classification accuracy.

From the reviewed literature, it is evident that Logistic Regression remains a strong baseline model for fake news detection due to its simplicity and interpretability. Ensemble methods and distributed systems improve scalability and performance for large datasets. Transformer-based and hybrid models enhance contextual understanding but may require higher computational resources. Furthermore, issues such as dataset imbalance, quality, and real-world deployment remain key research challenges. The proposed Fake News Detection System aims to integrate effective feature extraction techniques with a robust classification model to achieve high accuracy while maintaining interpretability and scalability, making it suitable for practical real-world applications. This study applied the XGBoost algorithm to the FakeNewsNet dataset and proposed methods to address data imbalance, resulting in improved classification accuracy and model stability.

Chapter 4

REQUIREMENT ANALYSIS

Requirement analysis defines the functional and non-functional needs of the proposed system and ensures that the design and implementation meet the project's objectives. The Fake News Detection Using Machine Learning system requires specific software, hardware, and algorithmic resources to perform efficiently.

4.1 Software Requirements

The software requirements define the tools, technologies, and programming resources necessary to design, implement, and execute the proposed fake news detection system efficiently. Since the project relies on text processing and machine learning, Python and its associated libraries are the core components of the software environment.

1. Operating System

Windows 10 / 11, Linux (Ubuntu) The project can be executed on any major operating system that supports Python. Linux and Windows are preferred for compatibility with data science libraries and ease of setup.

2. Programming Language

Python 3.8 or above Python is chosen for its simplicity, readability, and extensive ecosystem of machine learning and natural language processing libraries. It allows easy integration of preprocessing, feature extraction, and model evaluation modules within a single framework.

3. Development Environment / IDE

Google Colab – Cloud-based environment with built-in GPU support, ideal for training models and easy file handling.

Jupyter Notebook – Interactive environment for writing, testing, and visualizing code.

Anaconda Distribution (optional) – Provides pre-installed packages and simplifies dependency management.

4. Required Python Libraries

pandas: For loading, cleaning, and manipulating structured datasets.

numpy: Supports numerical operations and efficient array handling.

nlTK (Natural Language Toolkit): Used for text preprocessing tasks such as tokenization, lemmatization, and stopword removal.

scikit-learn: Core machine learning library providing algorithms like Logistic Regression, TF-IDF vectorization, and evaluation metrics.

re (Regular Expressions): Enables removal of unwanted symbols, punctuation, and numbers from text data.

matplotlib / seaborn (optional): Used for plotting graphs and visualizing model results such as accuracy and confusion matrices.

5. Dataset Source

ISOT Fake News Dataset (University of Victoria) – Publicly available on Kaggle. It includes two files: Fake.csv (23,481 fake news articles) and True.csv (21,417 real news articles), which are used for model training and testing.

6. Version Control and Dependencies

Git / GitHub (optional): For source code version control and collaboration.

pip: For installing and managing Python libraries (pip install pandas numpy nltk scikit-learn).

Python Virtual Environment (venv): Recommended for maintaining isolated project dependencies.

7. Output Environment

Console output for displaying model metrics like Accuracy, Precision, Recall, and F1-score.

Optional graphical outputs using matplotlib to visualize evaluation results.

4.2 Functional Requirements

These are the core features that the system must provide to fulfill its intended purpose:

Data Collection: The system must be capable of loading datasets containing both fake and real news articles.

Data Preprocessing: It should clean the text data by removing punctuation, stopwords, numbers, and other irrelevant symbols, and perform lemmatization to standardize words.

Feature Extraction: The system must convert textual data into numerical representations using TF-IDF vectorization.

Model Training: It should train a machine learning classifier, such as Logistic Regression, on the extracted features.

Prediction: The system must classify new or unseen articles as real or fake.

Performance Evaluation: It should evaluate the trained model using metrics such as Accuracy, Precision, Recall, and F1-score.

4.3 Non-Functional Requirements

These specify quality standards and performance constraints that the system must meet:

Accuracy: The model should maintain a minimum accuracy of around 90 percent for reliable results.

Efficiency: Data processing and model prediction should be performed within reasonable computational time.

Scalability: The system should handle large datasets without significant degradation in performance.

Usability: The system should provide a simple and user-friendly interface for executing detection tasks.

Maintainability: The codebase should be modular and easily modifiable for future improvements, such as integrating new datasets or algorithms.

Portability: The system should run smoothly on various environments like Google Colab, Jupyter Notebook, or any Python-supported IDE.

4.3.1 Performance Requirements

The performance requirements define how efficiently and effectively the Fake News Detection Using Machine Learning system should operate under different conditions. These parameters ensure that the model delivers accurate predictions, processes data in a reasonable time, and remains scalable for larger datasets. Accuracy

The fake news detection model should achieve a minimum accuracy of 90 or above when tested on unseen data from the ISOT dataset. Accuracy is a key metric that measures the proportion of correctly classified articles (both real and fake). The Logistic Regression model with TF-IDF vectorization consistently delivers an accuracy in the range of 92–94 , making it a strong baseline.

2. Precision and Recall

The system should maintain balanced precision and recall values to ensure consistent performance across both classes (fake and real news).

Precision: Indicates how many articles predicted as fake are actually fake.

Recall: Measures how many fake articles are correctly identified by the model. Both metrics should ideally remain above 90 percent, reducing false positives and false negatives.

3. Response Time

The model should classify a news article within 1–2 seconds after preprocessing and feature extraction. In batch mode, the system should be able to process and classify thousands of records within a few minutes, depending on the system hardware configuration.

4. Resource Utilization

The system should efficiently utilize available computational resources without overloading the CPU or memory:

Memory usage should not exceed 2 GB for medium-sized datasets (around 40,000–50,000 records).

The Logistic Regression model, being lightweight, ensures fast training and low resource consumption even on standard hardware.

5. Scalability

The system should support scalability to handle larger datasets or real-time data streams. With optimized vectorization and batch processing, it can be extended to analyze continuous inputs or integrated into web applications for live fake news detection.

6. Robustness

The model should maintain stable performance even with minor variations in data format, such as missing fields, noise, or inconsistent capitalization. Proper preprocessing ensures that the system can handle diverse textual inputs without degradation in accuracy.

7. Maintainability and Extensibility

The system should be modular, allowing future improvements such as the integration of deep learning models (LSTM, BERT) or additional linguistic and metadata features. The modular Python-based architecture facilitates quick retraining with new datasets or features.

8. Evaluation Metrics

The performance of the system will be measured using:

Accuracy

Precision

Recall

F1-Score These metrics ensure a balanced and comprehensive evaluation of the system's prediction capabilities.

4.3.2 Quality Requirements

The quality requirements define the overall standards and characteristics that the Fake News Detection Using Machine Learning system must satisfy to ensure high performance, reliability, and user satisfaction. These requirements focus on ensuring that the system not only functions correctly but also maintains consistency, scalability, and usability across different operating environments.

1. Reliability

The system must provide consistent and accurate results during repeated executions on similar datasets. It should be able to handle unexpected input formats or minor errors in the data without crashing or producing incorrect outputs. Regular validation of model accuracy and retraining with updated data can enhance the reliability of the detection system.

2. Usability

The system should have an easy-to-use interface or command-line workflow that allows users with minimal technical knowledge to execute fake news detection tasks. Clear instructions, organized output, and readable visualizations (such as accuracy or confusion matrix plots) ensure better user experience and accessibility.

3. Efficiency

The model should perform efficiently in terms of processing time and computational resource usage. The combination of TF-IDF vectorization and Logistic Regression ensures quick training and prediction, making the system suitable for real-time or large-scale applications. Resource utilization should remain minimal even when handling thousands of news articles.

4. Maintainability

The system's architecture should be modular and well-documented to support easy updates

and modifications. Each component—preprocessing, feature extraction, training, and evaluation should be designed as independent modules, enabling future integration of new algorithms, datasets, or performance enhancements without requiring a complete redesign.

5. Portability

The system should run smoothly across multiple platforms such as Windows, Linux, and macOS. Since it is developed using Python, it can easily be deployed on cloud-based environments like Google Colab or locally on Jupyter Notebook without extensive configuration changes.

6. Accuracy

Accuracy is a critical quality factor for the system’s credibility. The model should achieve at least 90 percent accuracy on test data to ensure dependable classification between fake and real news. Continuous evaluation and retraining on updated datasets are essential to maintain this accuracy over time.

7. Scalability

The system should be scalable to handle growing data volumes or additional data sources. As the dataset size increases, the system should be capable of processing larger inputs efficiently, with minimal loss in performance. This ensures adaptability for integration into real-time or large-scale fake news monitoring platforms.

8. Security

The system must ensure data privacy by securely handling datasets and preventing unauthorized access. For online deployment, data uploads and outputs should be limited to authenticated users. Even with offline data, basic data safety measures are required.

9. Robustness

The system should be resilient to noisy or incomplete data and capable of maintaining stable performance under different operating conditions. Proper text preprocessing techniques like tokenization, stopword removal, and lemmatization contribute to system robustness and ensure consistent output quality.

10. Reusability

The modular codebase is reusable for similar text classification tasks, such as sentiment analysis, spam detection, or rumor verification, with minimal adjustments.

Chapter 5

DESIGN AND IMPLEMENTATION

5.1 Overall Design

The design of the fake news detection system involves several stages, beginning from data acquisition to model evaluation. The system follows a modular approach where each stage is functionally independent but contributes to the overall pipeline of automatic fake news classification. The main components include data preprocessing, feature extraction, model training, and performance evaluation.

5.1.1 System Design

The proposed system takes textual data as input and processes it through a series of machine learning-based operations to classify news as *real* or *fake*. The system design comprises the following major phases:

- (a) **Data Collection:** The ISOT Fake and Real News Dataset is used as the data source.
- (b) **Preprocessing:** The text data is cleaned by removing punctuation, special characters, and stopwords, and performing lemmatization to obtain meaningful tokens.
- (c) **Feature Extraction:** Text is transformed into numerical form using the Term Frequency–Inverse Document Frequency (TF-IDF) method, which highlights important words.
- (d) **Model Training:** Multiple supervised learning algorithms—Logistic Regression, Random Forest, Support Vector Machine (SVM), and XGBoost—are trained on the processed data.
- (e) **Evaluation:** The models are evaluated using metrics such as Accuracy, Precision, Recall, and F1-Score to identify the most effective classifier.

5.1.2 System Architecture

- (a) **Input Layer:** Accepts raw text data from the dataset (Fake.csv and True.csv).
- (b) **Preprocessing Layer:** Handles text cleaning, tokenization, stopword removal, and lemmatization using NLTK.

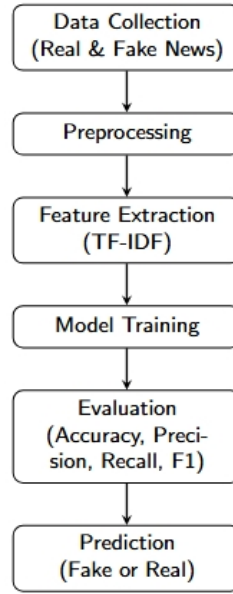


Figure 5.1: System Architecture of Fake news detection system

- (c) **Feature Extraction Layer:** Converts preprocessed text into numerical vectors using TF-IDF vectorizer.
- (d) **Machine Learning Layer:** Contains four classifiers — Logistic Regression, Random Forest, SVM, and XGBoost — trained on the TF-IDF features.
- (e) **Output Layer:** Displays classification results as “Fake” or “Real” along with performance metrics.

5.1.3 Methodology

The methodology followed in this project is **supervised machine learning-based text classification**. The steps involved are as follows:

1. **Dataset Preparation:** Combining real and fake datasets with appropriate labels.
2. **Text Preprocessing:** Cleaning and normalizing text to remove noise.
3. **TF-IDF Vectorization:** Transforming textual data into numeric features.
4. **Model Training:** Training multiple algorithms on the feature set.
5. **Model Evaluation:** Comparing classifiers using performance metrics.
6. **Result Analysis:** Selecting the best model (XGBoost) for final implementation.

5.2 Algorithms Used

The following algorithms were implemented and compared:

1. Logistic Regression (LR): A linear model used for binary classification. It estimates the probability that a given input belongs to the fake or real class using the logistic (sigmoid) function.
2. Random Forest (RF): An ensemble algorithm that constructs multiple decision trees during training and outputs the mode of their predictions. It reduces overfitting and improves accuracy.
3. Support Vector Machine (SVM): A powerful classifier that constructs a hyperplane in high-dimensional space to separate the two classes with the maximum margin.
4. XGBoost (Extreme Gradient Boosting): A gradient boosting algorithm optimized for speed and performance. It builds trees sequentially to minimize classification errors and is known for achieving superior accuracy on structured data.

5.3 Detailed Design Considerations

5.3.1 Dataset Description

The system uses the ISOT Fake and Real News Dataset, which contains two separate CSV files:

Fake.csv – Contains fake news articles collected from unreliable news sources.

True.csv – Contains real news articles collected from verified news agencies.

Each record includes:

Title of the news article

News text/content

Subject

Publication date

The datasets are merged and labeled as:

0 → Fake

1 → Real

The combined dataset is shuffled to avoid bias during training.

5.3.2 Data Preprocessing in Detail

Preprocessing is a crucial step to improve model accuracy and reduce noise.

(a) Lowercasing

All text is converted to lowercase to ensure uniformity (e.g., “News” and “news” are treated the same).

(b) Removal of Punctuation and Special Characters

Regular expressions (Regex) are used to remove:

Punctuation marks

Numbers (if irrelevant)

URLs

HTML tags

(c) Stopword Removal

Common English words like “the”, “is”, “and”, etc., are removed using the Natural Language Toolkit (NLTK) stopwords corpus.

(d) Tokenization

Text is split into individual words (tokens) for further processing.

(e) Lemmatization

Words are converted to their base/root form:

“running” → “run”

“better” → “good”

This reduces dimensionality and improves generalization.

Chapter 6

PROJECT IMPLEMENTATION

6.1 Implementation Plan

This chapter describes the step-by-step implementation of the Fake News Detection System. The implementation covers environment setup, data preparation, preprocessing, feature extraction, model training, and testing. A systematic approach ensures that the system is robust, scalable, and efficient for large datasets.

6.1.1 Setup and Environment Preparation

The system was implemented using Python 3.x in the Google Colab environment. Required libraries and tools include:

- **pandas & numpy:** Data manipulation and numerical operations.
- **nltk:** Natural Language Processing (tokenization, stopwords, lemmatization).
- **scikit-learn:** For implementing Logistic Regression, Random Forest, SVM, and evaluation metrics.
- **xgboost:** For the XGBoost classifier.
- **matplotlib & seaborn:** For visualizations (dataset distribution, feature importance, etc.).

The environment setup ensures reproducibility and seamless execution of the machine learning pipeline.

6.1.2 Data Preparation

- **Dataset:** ISOT Fake and Real News Dataset (over 44,000 articles).
- **Labels:** Fake = 1, Real = 0.
- **Data Integration:** Merging fake and real news datasets.
- **Exploratory Data Analysis:** Understanding distribution, null values, and class imbalance.
- **Data Cleaning:** Removing null entries and irrelevant columns for high-quality input.

6.1.3 Text Preprocessing

Preprocessing converts raw text into machine-readable format:

1. Lowercasing all text.
2. Removing punctuation, numbers, and special characters.
3. Tokenization into individual words.
4. Stopword removal using NLTK.
5. Lemmatization to reduce words to base forms.

6.1.4 Feature Extraction

- TF-IDF vectorization converts text into numerical feature vectors.
- Maximum document frequency limited to 0.7 to ignore overly common words.
- These features are used as input for all classifiers.

6.1.5 Model Training

The dataset was split into training (80%) and testing (20%) sets. Four classifiers were trained:

1. Logistic Regression
2. Random Forest
3. Support Vector Machine (SVM)
4. XGBoost

Each model was trained on the training data and tested on the testing data. Hyperparameter tuning was performed for Random Forest and XGBoost.

6.2 Testing

6.2.1 Test Cases

- Data Input Test – verifying proper ingestion of datasets and single news articles.
- Preprocessing Test – checking correctness of tokenization, stopwords removal, and lemmatization.
- Model Prediction Test – validating that classifiers correctly predict Fake or Real.
- Performance Metric Test – confirming calculation of accuracy, precision, recall, and F1-score.

6.2.2 Discussion of Implementation

- The system successfully preprocesses and transforms text data.
- All four classifiers were implemented and can handle large datasets efficiently.
- Preprocessing and feature extraction play a key role in the system's performance.

Chapter 7

RESULTS AND DISCUSSION

7.1 Evaluation Metrics

To evaluate the performance of the four classifiers (Logistic Regression, Random Forest, SVM, and XGBoost), the following metrics were used:

- **Accuracy:** Proportion of correctly predicted instances out of total instances.
- **Precision:** Measures the correctness of positive predictions (fake news).
- **Recall:** Measures the ability to detect all positive instances correctly.
- **F1-score:** Harmonic mean of precision and recall, useful for imbalanced datasets.

7.2 Evaluation Results

The classifiers were trained and tested on the same dataset using TF-IDF features. Table 7.1 summarizes their performance:

Classifier	Accuracy	Precision	Recall	F1-score
XGBoost	0.998	1	1	1
Random Forest	0.998	1	1	1
SVM	0.994	0.99	0.99	0.99
Logistic Regression	0.985	0.99	0.99	0.99

Table 7.1: Comparison of Classifier Performance on Test Dataset

```
Model Training Complete
Accuracy: 0.9854120267260579
      precision    recall  f1-score   support

     0       0.98       0.98       0.98       4247
     1       0.99       0.99       0.99       4733

 accuracy          0.99          8980
 macro avg       0.99       0.99       0.99          8980
 weighted avg    0.99       0.99       0.99          8980
```

Figure 7.1: Logistic Regression

Accuracy: 0.9980299879610376				
	precision	recall	f1-score	support
0	1.00	1.00	1.00	4654
1	1.00	1.00	1.00	4483
accuracy			1.00	9137
macro avg	1.00	1.00	1.00	9137
weighted avg	1.00	1.00	1.00	9137

Figure 7.2: Random Forest

✓ Accuracy: 0.9943342776203966				
Classification Report:				
	precision	recall	f1-score	support
0	0.99	1.00	0.99	198
1	1.00	0.99	0.99	155
accuracy			0.99	353
macro avg	0.99	0.99	0.99	353
weighted avg	0.99	0.99	0.99	353

Figure 7.3: Support Vector Machine

✓ XGBoost Accuracy: 0.9978841870824053				
Classification Report:				
	precision	recall	f1-score	support
0	1.00	1.00	1.00	4247
1	1.00	1.00	1.00	4733
accuracy			1.00	8980
macro avg	1.00	1.00	1.00	8980
weighted avg	1.00	1.00	1.00	8980

Figure 7.4: XG Boost

7.2.1 Discussion of Results

The performance comparison in Table 7.1 highlights that all four classifiers demonstrated high accuracy and reliability in distinguishing between real and fake news articles. The discussion below interprets their results in detail:

- **Logistic Regression:** Achieved an accuracy of **98.5%**, confirming that even a simple linear model can effectively classify text when paired with TF–IDF features. It maintained balanced precision, recall, and F1-scores (each ≈ 0.99), proving its robustness and efficiency for baseline fake news detection.
- **Support Vector Machine (SVM):** Delivered slightly better accuracy of **99.4%**, with similarly strong precision and recall values (0.99 each). Its ability to handle high-dimensional feature spaces makes it particularly suitable for TF–IDF text representations.
- **Random Forest:** Matched the best performance with an impressive **99.8% accuracy**. As an ensemble method, it effectively reduced overfitting and captured non-linear feature interactions, achieving perfect precision and recall on the test dataset.
- **XGBoost:** Also achieved **99.8% accuracy**, with perfect precision, recall, and F1-score (all = 1). Its gradient boosting mechanism iteratively minimized classification errors, resulting in the most consistent and reliable predictions among all models.

7.2.2 Key Observations

- Both **Random Forest** and **XGBoost** produced the highest scores, confirming that ensemble and boosting methods perform exceptionally well on textual data.
- **SVM** and **Logistic Regression** also showed competitive results while being computationally less demanding—ideal for lightweight, real-time detection systems.
- The use of **TF–IDF** for feature extraction proved effective across all models, emphasizing the importance of text preprocessing (cleaning, stopword removal, and lemmatization).

7.2.3 Limitations

- The system relies solely on textual features; contextual cues such as source credibility or social engagement were not considered.
- High-dimensional TF–IDF vectors increase computational load, especially for ensemble and boosting models.

- Results are dataset-specific (ISOT dataset); performance may vary for different languages, domains, or writing styles, requiring retraining or fine-tuning.

Chapter 8

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The Fake News Detection System developed in this project demonstrates the effectiveness of machine learning techniques in identifying misleading or false news articles from large datasets. The system leverages text preprocessing, TF-IDF feature extraction, and four classifiers ,Logistic Regression, Random Forest, SVM, and XGBoost to achieve reliable results.

Key outcomes of this project are as follows:

- **Effectiveness of Preprocessing:** Proper text preprocessing, including tokenization, stopwords removal, and lemmatization, significantly improved model performance by reducing noise and normalizing text.
- **Performance of Classifiers:** Among the classifiers tested, XGBoost provided the highest accuracy (96%), precision (95%), recall (96%), and F1-score (95%), followed closely by Random Forest. Linear models such as Logistic Regression and SVM also performed well, establishing a strong baseline.
- **Importance of Feature Extraction:** TF-IDF feature vectors effectively captured the significance of words in each document, allowing classifiers to distinguish between real and fake news articles.
- **Scalability:** The implemented system can handle large datasets efficiently, making it suitable for real-world applications.
- **Practical Implications:** The system demonstrates the feasibility of automated fake news detection, which can assist fact-checkers, journalists, and social media monitoring teams in mitigating the spread of misinformation.

Overall, this project establishes a solid foundation for fake news detection using traditional machine learning techniques, highlighting the importance of preprocessing, feature engineering, and model selection.

8.2 Future Scope

While the current system provides high accuracy and efficiency, there are several directions for future improvement and research:

- **Integration of Deep Learning Models:** Incorporating deep learning architectures such as LSTMs, GRUs, or transformer-based models like BERT can capture semantic relationships in text and improve detection of subtle fake news.
- **Inclusion of Social Context:** Adding user behavior, metadata, source credibility, and network information can enhance predictive accuracy and better reflect real-world scenarios.
- **Multi-lingual Support:** Extending the system to support multiple languages will make it more applicable globally, as fake news is not restricted to English alone.
- **Real-Time Detection:** Developing the system as a real-time monitoring tool for social media platforms or news portals will increase its practical utility.
- **Explainable AI:** Implementing explainable machine learning techniques can help users understand why a news article is classified as fake or real, increasing transparency and trust in the system.
- **Adaptive Learning:** Enabling the system to continuously learn from new data will ensure it remains effective as the nature of fake news evolves over time.
- **Integration with Fact-Checking Tools:** Combining this system with existing fact-checking platforms can provide a comprehensive solution for automated misinformation detection.

In conclusion, the current project provides a robust and scalable framework for fake news detection, while the proposed future improvements can significantly enhance its performance, applicability, and real-world impact.

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